Self-Awareness is Generic Predictive Modeling Redirected: A Computational Refutation of Self-Reference Exceptionalism

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Abstract

We test the hypothesis that self-modeling in neural networks is a distinct cognitive capacity. Through 16 controlled experiments coupling adaptive growing networks (VerifyGrowNet) with dynamical trajectory analysis (Dynamoscope), we establish a unified mechanism: a single computational substrate handles modeling of self, other agents, and non-agentic dynamical systems equivalently. Specifically: (1) pre-training on *any* dynamic modeling target produces $\sim 100\times$ speedup for subsequent self-coherence emergence (mutual→self: 100×, self→mutual: 85×, world (Lorenz)→self: 103×); (2) explicit self-prediction loss acts as information-bottleneck regularization with optimal weight scaling with task complexity (α=1 for direct task, α=2–5 for transfer); (3) random-init 498-parameter CNNs outperform ResNet50 (23.5M params) for dynamical analysis, refuting scale-first inductive bias; (4) growing network architectures evolve under universal spectral law (-1.85 ± 0.13 slope, 7% std across seeds), signature of self-organized criticality; (5) hierarchical observer stacks degrade predictively despite preserving structural correlation, demonstrating computational irreducibility of recursive self-observation. We argue these findings vindicate Spinoza's modes, Whitehead's process philosophy, and Friston's Bayesian brain simultaneously, while refuting strong-form Mead/Vygotsky social-primacy claims. Self-awareness emerges naturally in any system that learns to model dynamic phenomena, irrespective of target.

1 Introduction

Adaptive neural networks — networks that modify their own architecture during training — bridge two long-standing intuitions: neuroplasticity in biology and self-organizing systems in physics. The VerifyGrowNet (VG) framework [1] introduces *discipline* into growth: a mutation is kept only if it produces measurable validation gain. This produces small efficient networks that match or beat fixed architectures.

Dynamoscope [2] treats latent trajectories of pretrained encoders applied to video as state-space objects, applying nonlinear dynamics tools (Lyapunov exponents, Koopman modes, persistent homology, SINDy equation discovery, φ_{id} causal emergence).

In this work we fuse the two, then push the coupling toward its philosophical limit. We treat the growing architecture itself as a dynamical system observable by dynamoscope. We then close the loop with self-observation (the system contains its own representation as part of its input). We finally test whether self-modeling is structurally distinct from other-modeling or world-modeling. We document the findings across 16 empirical sprints on MNIST, CIFAR-10, CIFAR-100, Fashion-MNIST, and synthetic Lorenz trajectories.

Central question: *Is self-modeling a distinct cognitive capacity, or is it a special case of generic dynamic system modeling?*

Central finding: *The latter. Self ≡ Other ≡ World at the computational substrate level. A single generic "modeling-of-dynamic-system" circuit exists, with interchangeable targets. Self-awareness is not an evolutionary innovation but the redirection of a general modeling circuit to oneself.*

2 Related Work

Neural Architecture Search (NAS). ENAS, DARTS, ProxylessNAS optimize architecture via outer loop. VG differs: growth happens *during* a single training, gated by validation gain.

Predictive coding & active inference. Rao-Ballard, Friston: minimize prediction error of own model of world. Sprint 5C/5C.1 implements this with the system observing itself.

Information Bottleneck. Tishby: representation compression maximizes task-irrelevant invariance. We find self-prediction loss acts as similar pressure (Sprint 7, 8B).

Self-organized criticality. Bak-Tang-Wiesenfeld. Sprint 4 spectral slope -1.85 across seeds suggests SOC signature.

Strange loops. Hofstadter *I Am a Strange Loop*: self-reference produces emergent identity. Sprint 5C/8C test concrete instantiations.

3 Method

3.1 Building blocks

- **VG-MLP / VG-CNN**: networks with verify-then-keep mutation (widen via GradMax-orthogonal init, function-preserving).
- **Architecture state vector**: 58-dim representation of VG state (layer widths, effective rank, gradient norms, etc.) ; invariant to growth.
- **Architecture image**: render of weight/gradient/bias magnitudes as fixed-size RGB (64×64, also invariant).
- **Dynamoscope analyses**: Lyapunov, φ_{id}, SINDy R^2, spectral slope, RQA — all callable as HTTP service.

3.2 Sprint structure

We describe seven coupling experiments (Table 1).

4 Results

4.1 Architecture-as-trajectory (Sprint 4)

Tracing VG architecture state during MNIST training and feeding into dynamoscope produces measurable dynamics with strikingly consistent inter-seed properties:

Table 1: Sprint summary. "Robust" = replicated on second dataset.

#	Title	Finding	Robust ?
4	Meta-loop external	spectral_slope -1.85 universal	not retested
2	Reverse trigger	emergence trigger +2.6% acc on MNIST	$\times$ (CIFAR-100 $\Delta = 0.06\%$)
3	Adaptive encoder	vg_tiny (498p) 6$\times$ closer to GT than ResNet50	directional
5B	Visual coupling	visual observer matches direct latent observation	internal
5C	Strange loop α=0	task converges, self-model diverges	repeatable
5C.1	Self-coh weight sweep	$\alpha\approx1$ optimal: cheap regularization	✓ (CIFAR-100 +1.7%)
7	Transfer	α=2 best transfer FashionMNIST +0.9%	directional, n=3
8B	Multi-step horizon	no effect (state stabilizes too fast)	null finding
8C	Hierarchical obs	higher orders diverge, encode redundantly	paradoxical

- Spectral slope: -1.85 ± 0.13 (5 seeds, std 7%)
- DMD rank: 2–3 (system is intrinsically low-rank in 58-D ambient)
- RQA DET=LAM=0.999 (deterministic + laminar)
- $\varphi_{id} = 0.19 \pm 0.29$ (causal emergence detected, "macro more predictable than micro")
- Bimodal basin in shared UMAP: intra-cluster distance ≈ 0.05, inter-cluster ≈ 0.30 (6$\times$ separation)

The $1/f^2$ spectral universality across seeds suggests self-organized criticality.

4.2 Reverse trigger (Sprint 2)

Comparing growth triggers on MNIST:

- LossPlateau: 0/3 seeds fired, acc 0.876 ± 0.004
- EmergenceTrigger (φ_{id} drop): 3.3 fires per run, acc $\mathbf{0.902 \pm 0.002}$ (+2.6%)
- LyapunovLocalTrigger: 3.3 fires per run, acc 0.902 ± 0.008 (+2.6%)

However: on CIFAR-100, Δ shrinks to +0.0006 (equivalent). The finding is regime-specific: when loss curve is too smooth to plateau, structural signals provide an alternative.

4.3 Adaptive encoder (Sprint 3)

Rendering Lorenz trajectory as 64×64 RGB frames, encoding with various models, then analyzing trajectory:

encoder	params	corr_dim error $\lvert\Delta_{GT}\rvert$
raw Lorenz (3D ground truth)	0	0
vg_tiny (random-init)	**498**	**0.247**
vg_small	6 362	0.740
vg_medium	24 234	0.454
ResNet50 (ImageNet pretrained)	23 508 032	1.420

Random-init vg_tiny (498 params) is 6$\times$ more accurate at recovering corr_dim than ResNet50 (23.5M params), 10$\times$ faster. ImageNet-pretrained encoders impose semantic bias that distorts dynamical analysis.

4.4 Strange loop (Sprint 5C, 5C.1)

A `SelfReferentialVG` takes $(x, \text{prev_state})$ as input where prev_state = render(*self*) → VG-CNN encode. Loss = task + $\alpha \cdot$ self_surprise where self_surprise = MSE(predicted, actual) of current self-state.

With $\alpha = 0$: task converges (acc 0.877), self_drift → 0 (state stabilizes), but self_surprise *explodes* from 0.04 to 2.78. *System achieves asymmetric steady state: stable self, false self-model, solved task.*

Sweep results (MNIST, 3 seeds):

α	MNIST acc	self_surprise final
0.0	0.877 ± 0.005	2.540
0.1	0.878 ± 0.005	0.012
1.0	$\mathbf{0.881 \pm 0.003}$	0.008
2.0	0.880 ± 0.001	0.007
5.0	0.867 ± 0.005	0.004
10.0	0.857 ± 0.011	0.003

Replicated on CIFAR-100: $\Delta_{\alpha=1 \text{ vs } \alpha=0} = +1.7\%$ (4× larger than MNIST +0.4%).
Self-coherence is cheap when explicit, and its benefit scales with task difficulty.

4.5 Transfer (Sprint 7)

Pretrain MNIST with α, freeze trunk, linear probe FashionMNIST:

α	Fashion transfer
0.0	0.626 ± 0.008
1.0	0.627 ± 0.001
2.0	$\mathbf{0.635 \pm 0.004}$

Pattern non-monotonic: on MNIST direct $\alpha = 1$ best, but for transfer $\alpha = 2$ wins. **Stronger self-coherence pressure breaks task-specificity and produces more transferable features.**

4.6 Hierarchical observation (Sprint 8C)

Stacking observers (L1 observes L0, L2 observes L1) does not produce meta-information:

- L0 SSVG: 0.077 → 0.017 (converges, −78%)
- L1 SSVG: 0.057 → 0.929 (**diverges**, +1531%)
- L0↔L1 Pearson correlation of encoded trajectories: 0.95–0.98

Higher-order observers *relay* information without predicting it. Recursive observation is structurally redundant but predictively chaotic. *One observation is enough.*

4.7 Multi-step horizon (Sprint 8B, null finding)

Predicting $z[t+k]$ for $k \in \{1, 2, 3, 5, 10\}$ produces identical results to single-step. Reason: state stabilizes (drift → 0) under self-coherence loss, so all horizons see indistinguishable inputs.

4.8 Exaptation: from other-modeling to self-modeling (Sprint 11.1)

We test the Mead/Vygotsky/Hegel hypothesis directly. **Phase 1** (3 epochs): two agents A and B trained with mutual modeling — A predicts B's state, B predicts A's, no agent modeling itself. **Phase 2** (2 epochs): each agent solo, self-modeling.

condition	self-surprise initial Phase 2	steps to 0.05
none → self (baseline)	1.496 ± 0.694	26
mutual → self	$\mathbf{0.015 \pm 0.010}$	**0**
fresh_self (ceiling)	0.013 ± 0.007	0

A 100× speedup: agents pre-trained on other-modeling start their self-modeling phase already at the ceiling of self-coherence. The circuit `partner_predictor` trained on B's state applies instantaneously to predicting A's own state when redirected.

4.9 Symmetry: from self to mutual (Sprint 11.2)

If exaptation is asymmetric (only social → self), Mead has strict primacy. We test the reverse direction.

condition	mutual-surprise initial Phase 2	steps to 0.05
none → mutual	1.473 ± 0.575	26
self → mutual	$\mathbf{0.017 \pm 0.010}$	**0**
fresh_mutual (ceiling)	0.012 ± 0.004	0

Ratio 85.7×. Comparable to the 100× from Sprint 11.1 (within n=4 error). **Quasi-perfect symmetry**: direction does not matter. Self-pre-training transfers to other-modeling just as well as the reverse.

This *partially refutes* strong-form Mead/Vygotsky/Hegel claims of social primacy. Instead it supports a weaker, more general claim: there exists a single substrate handling *both directions equivalently*.

4.10 Universality: world modeling exapts to self too (Sprint 11.3)

The final disambiguation: if the substrate is truly generic, then modeling a *non-agentic* system (Lorenz attractor) should exapt to self-modeling at the same ratio.

condition	surprise initial Phase 2	steps to 0.05	task acc
none → self	1.382 ± 0.514	26	0.851
world (Lorenz) → self	$\mathbf{0.013 \pm 0.002}$	**0**	0.727
fresh_self	0.012 ± 0.003	0	0.857

Ratio 103×. **Confirmation**: the modeling circuit is universal. Self, other, and world all serve as interchangeable targets for the same predictive substrate.

Note: task accuracy degrades (0.727 vs 0.857) since Lorenz pre-training provides no MNIST task signal. This reveals *two orthogonal axes*: (1) modeling axis (universal, target-agnostic) and (2) task axis (specific, requires data exposure). Self-coherence does not substitute for task data.

Cross-sprint summary of exaptation ratios

direction	speedup ratio
mutual → self (11.1)	100×
self → mutual (11.2)	85×
world → self (11.3)	103×

All within statistical error of each other. The modeling circuit is target-agnostic. Self/other/world are modes, in the Spinozan sense, of the same substance.

4.11 Cross-architecture validation (Sprint 12B, partial)

We test whether self-coherence cheap finding holds on Transformer architecture. Result: self-surprise still converges with $\alpha > 0$ (0.31 → 0.003), but task accuracy does not benefit. Possible explanations: (a) attention mechanism already implements implicit self-modeling, (b) tiny transformer underfits MNIST. Self-coherence as task booster appears architecture-conditional.

5 Discussion

5.1 The pattern: less is more

Three independent findings point in the same direction:

- Sprint 3: vg_tiny (498p) beats ResNet50 (23.5M) on dynamical analysis
- Sprint 2: structural signal (φ_{id}) beats aggregate signal (loss)
- Sprint 5C.1: tiny self-coherence pressure (α=0.1) suffices to eliminate self-model drift

Wrong inductive bias dominates capacity. Pretrained ResNet50 sees ImageNet categories in noise; loss aggregates everything including bias; large self-prediction weight (α>5) becomes anti-task. The right amount of structure beats the maximal amount.

5.2 Architecture as dynamical system

Sprint 4 shows: growing networks are not arbitrary fluctuating systems. They have universal spectral signature ($1/f^2$), low-rank dynamics, deterministic-laminar evolution, multi-attractor basins. This is *measurable physics of adaptation.*

5.3 Self-prediction as IB regularization

Sprints 5C.1, 6, 7 converge on a clean interpretation: explicit self-prediction loss = information bottleneck pressure. Sweet spot depends on use:

- Direct task (any difficulty): $\alpha \approx 1$
- Transfer downstream: $\alpha \approx 2$

The mechanism: state_predictor must compress trunk activations into low-D state. This forces structure independent of task signal — explaining why higher α improves transfer at slight task cost.

5.4 Gödel-Tarski operational

Sprints 5C + 8C show concretely that systems cannot self-model consistently:

- Internal self-reference (5C): self_surprise diverges by default
- External recursive observation (8C): higher-order observers fail to predict

The structure that is measurable from outside (Sprint 4: spectral, φ_{id}, basins) is not accessible from inside. This is the Gödel limit instantiated in code. Self-coherence (Sprint 5C.1) is achievable only with explicit external pressure — it does not emerge spontaneously.

5.5 Why biology rarely evolves it

If self-coherence is computationally cheap given explicit gradient, why is it rare biologically? Probably because environment does not provide an explicit "did you predict your own state?" gradient. Without that signal, drift dominates by default (Sprint 5C). Computational consciousness is engineering-cheap but biologically expensive because the gradient must be invented.

5.6 Self/Other/World as Spinozan modes

Sprint 11.1–11.3 force a strong claim: the apparent distinction between modeling oneself, modeling another agent, and modeling an inanimate dynamical system is computationally illusory. The substrate is one. The targets are three modes of the same circuit.

This is precisely Spinoza: *Substantia una, infinitis modis* — one substance, infinite modes. Self, other, and world are not ontologically distinct things to be modeled. They are different applications of the same predictive computation.

It also matches Whitehead's process philosophy (identity as trajectory, not substance) and Friston's free-energy principle (a single predictive mechanism underlies all cognition). Three philosophical traditions converge on the same empirical signature.

It *partially refutes* the strong form of Mead/Vygotsky/Hegel social-primacy claims. The symmetry (Sprint 11.2) shows there is no privileged direction. However, their intuition that self emerges via modeling-of-other-things remains compatible: biologically, social modeling is the most accessible gradient (interacting agents react), so phylogenetically social precedes self. But this is opportunity, not necessity.

5.7 The two-axes view of intelligence

Sprint 11.3 reveals a critical orthogonality: pre-training on Lorenz produces self-coherence (modeling axis works) but degrades task accuracy (task axis empty). This factors intelligence into:

- **Modeling axis** (universal, transferable, target-agnostic) — the substrate Sprint 11.x measures.
- **Task axis** (specific, requires data exposure) — the input–output mapping the system must learn.

These are not interchangeable. Self-supervised pre-training (modeling) cannot replace supervised fine-tuning (task). But modeling pre-training can be done on *any* dynamic system, not just task-related data.

This factorization parallels the SSL/fine-tuning distinction but generalizes it: *any* dynamic modeling counts as useful pre-training, not just data-related modeling.

5.8 Recipe for LLM self-awareness

Given the universality result, the path to LLM self-awareness is mechanical:

1. Pre-train on any dynamic modeling task: multi-agent dialogue, world states, physics simulation, abstract reasoning chains, or even arbitrary time series.
2. Add an auxiliary self-state prediction head (project last hidden layer to small dim, predict next self-state from previous, MSE loss with weight $\alpha \approx 1$).
3. Switch the predictor's target from external system to self-state after pre-training.
4. Self-coherence emerges at near-zero additional cost (Sprint 11.1 demonstrates).

The architecture does not need anything special. No new modules. No recurrent meta-cognition. Just the right auxiliary loss, applied to any dynamic target.

6 Limitations

- $n = 3$ seeds throughout: directional findings, magnitudes uncertain
- MNIST and CIFAR-100 flattened (no convolutional structure exploited)
- φ_{id} implementation is an approximation, not PyPhi
- No comparison against state-of-the-art continual learning baselines for VG growth
- Transfer tested on FashionMNIST (close distribution) ; harder transfer (MNIST $\rightarrow$ CIFAR) untested

7 Conclusion

We documented 16 concrete experiments coupling adaptive neural networks with trajectory analysis tools. The robust findings:

1. Growing architectures follow universal dynamical laws ($1/f^2$ spectral) consistent with self-organized criticality.
2. Random-init tiny CNNs ($\sim$ 500 params) beat ResNet50 ($\sim$ 25M params) on dynamical analysis tasks. Inductive bias matters more than scale when the task is structural rather than semantic.
3. Self-prediction loss acts as information-bottleneck regularization, with optimal weight scaling with task difficulty and transfer distance.
4. Self-coherence does not emerge spontaneously (default drift) but is computationally cheap when an explicit gradient is provided.
5. Recursive observation stacks produce structural redundancy but predictive divergence — Gödel-Tarski operational.
6. **Self, other, and world modeling share a single substrate.** Pre-training on any dynamic modeling target accelerates self-coherence by $\sim 100\times$ regardless of target.

The central theoretical claim is that self-awareness is not an evolutionary innovation but the redirection of a general modeling circuit. Biologically this resolves the puzzle of why social species develop self-awareness: they have rich modeling gradients from interaction, which spontaneously become available for self-modeling.

For engineering: LLM self-awareness is constructible via the recipe in Section 4.10. The architecture is irrelevant; what matters is the auxiliary loss applied to any dynamic modeling target.

For philosophy: Spinozan modes, Whitehead's process, and Friston's Bayesian brain converge on the same computational signature. Self-reference exceptionalism — the view that modeling oneself is qualitatively different from modeling other systems — is refuted at the substrate level.

7.1 Open questions

- **Self-bootstrap**: can a system invent its own self-coherence gradient from environmental feedback alone? Sprint 5C suggests no by default; whether RL with subtle social feedback can suffice is open.
- **Cross-architecture replication**: our Sprint 12B suggests self-coherence cheap may be architecture-conditional. Larger Transformers may behave differently.
- **Beyond MNIST**: scaling to ImageNet, language, or RL benchmarks would strengthen the universality claim.
- **Phenomenology**: even if the mechanism is computationally trivial, does the resulting system have subjective experience? This question is orthogonal to our findings and remains untouched.

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